MPG No. 11.3.37 $4'/m$ [Type III, tetragonal] [T tensor]

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s).

$$\begin{bmatrix} T_{xx} & T_{xy} & 0 \\ T_{xy} & -T_{xx} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$T_{xx} = \mathbb{T}_{v}^{(1)}$
$T_{xy} = \mathbb{T}_{xy}^{(1)}$

* Rank 2 tensor (a). * Rank 3 tensor (ss). * Rank 3 tensor (sa). * Rank 3 tensor (a). * Rank 4 tensor (sss).

$$\begin{bmatrix} T_{xxxx} & 0 & T_{xxzz} & 0 & 0 & T_{xxxy} \\ 0 & -T_{xxxx} & -T_{xxzz} & 0 & 0 & T_{xxxy} \\ T_{xxzz} & -T_{xxzz} & 0 & 0 & 0 & T_{zzxy} \\ 0 & 0 & 0 & T_{yzyz} & T_{yzzx} & 0 \\ 0 & 0 & 0 & T_{yzzx} & -T_{yzyz} & 0 \\ T_{xxxy} & T_{xxxy} & T_{zzxy} & 0 & 0 & 0 \end{bmatrix}$$

$T_{xxxx} = -\mathbb{T}_{4v}^{(1)} + 2\mathbb{T}_{v}^{(1)} + 4\mathbb{T}_{v}^{(2)}$
$T_{xxzz} = \mathbb{T}_{4v}^{(1)} + \mathbb{T}_{v}^{(1)}$
$T_{xxxy} = -\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)}$
$T_{zzxy} = 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)}$
$T_{yzyz} = -\mathbb{T}_{4v}^{(1)} - \mathbb{T}_{v}^{(2)}$
$T_{yzzx} = 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(2)}$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & T_{xxyy} & T_{xxzz} & 0 & 0 & T_{xxxy} \\ -T_{xxyy} & 0 & -T_{xxzz} & 0 & 0 & T_{xxxy} \\ -T_{xxzz} & T_{xxzz} & 0 & 0 & 0 & T_{zzxy} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -T_{xxxy} & -T_{xxxy} & -T_{zzxy} & 0 & 0 & 0 \end{bmatrix}$$

$T_{xxyy} = 4\mathbb{M}_{xyz}^{(1)} - 2\mathbb{T}_{v}^{(3)}$
$T_{xxzz} = -4\mathbb{M}_{xyz}^{(1)} - \mathbb{T}_{v}^{(3)}$
$T_{xxxy} = 2\mathbb{M}_{z}^{\beta(1)} + \mathbb{T}_{xy}^{(3)}$
$T_{zzxy} = -4\mathbb{M}_{z}^{\beta(1)} + \mathbb{T}_{xy}^{(3)}$

* Rank 4 tensor (aas).

$$\begin{bmatrix} T_{yzyz} & T_{yzzx} & 0 \\ T_{yzzx} & -T_{yzyz} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$T_{yzyz} = 2\mathbb{T}_{v}^{(4)}$
$T_{yzzx} = 2\mathbb{T}_{xy}^{(4)}$

* Rank 4 tensor (aaa). * Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & 0 & T_{xxxy} \\ 0 & 0 & -T_{xxxy} \\ 0 & 0 & 0 \\ T_{yzyz} & T_{yzzx} & 0 \\ -T_{yzzx} & T_{yzyz} & 0 \\ 0 & 0 & T_{xyxy} \end{bmatrix}$$

$$T_{xxxy} = \mathbb{M}_{z}^{\beta(2)} - 2\mathbb{T}_{xy}^{(5)}$$
$$T_{yzyz} = \mathbb{M}_{xyz}^{(2)} - \mathbb{T}_{v}^{(5)}$$
$$T_{yzzx} = -\mathbb{M}_{z}^{\beta(2)} - \mathbb{T}_{xy}^{(5)}$$
$$T_{xyxy} = \mathbb{M}_{xyz}^{(2)} + 2\mathbb{T}_{v}^{(5)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & T_{yzyz} & T_{yzzx} & 0 \\ 0 & 0 & 0 & -T_{yzzx} & T_{yzyz} & 0 \\ T_{xyxx} & -T_{xyxx} & 0 & 0 & 0 & T_{xyxy} \end{bmatrix}$$

$$T_{yzyz} = \mathbb{M}_{xyz}^{(3)} - \mathbb{T}_{v}^{(6)}$$
$$T_{yzzx} = \mathbb{M}_{z}^{\beta(3)} + \mathbb{T}_{xy}^{(6)}$$
$$T_{xyxx} = \mathbb{M}_{z}^{\beta(3)} - 2\mathbb{T}_{xy}^{(6)}$$
$$T_{xyxy} = \mathbb{M}_{xyz}^{(3)} + 2\mathbb{T}_{v}^{(6)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} T_{xxxx} & T_{xxyy} & T_{xxzz} & 0 & 0 & T_{xxxy} & 0 & 0 & T_{xxyx} \\ -T_{xxyy} & -T_{xxxx} & -T_{xxzz} & 0 & 0 & T_{xxyx} & 0 & 0 & T_{xxxy} \\ T_{zzxx} & -T_{zzxx} & 0 & 0 & 0 & T_{zzxy} & 0 & 0 & T_{zzxy} \\ 0 & 0 & 0 & T_{yzyz} & T_{yzzx} & 0 & T_{yzzy} & T_{yzxz} & 0 \\ 0 & 0 & 0 & T_{yzxz} & -T_{yzzy} & 0 & T_{yzzx} & -T_{yzyz} & 0 \\ T_{xyxx} & T_{xyxx} & T_{xyzz} & 0 & 0 & T_{xyxy} & 0 & 0 & -T_{xyxy} \end{bmatrix}$$

$$T_{xxxx} = -\mathbb{T}_{4v}^{(1)} + 2\mathbb{T}_{v}^{(1)} + 4\mathbb{T}_{v}^{(2)}$$
$$T_{xxyy} = 4\mathbb{M}_{xyz}^{(1)} - 2\mathbb{T}_{v}^{(3)}$$
$$T_{xxzz} = -4\mathbb{M}_{xyz}^{(1)} + \mathbb{T}_{4v}^{(1)} + \mathbb{T}_{v}^{(1)} - \mathbb{T}_{v}^{(3)}$$
$$T_{xxxy} = 2\mathbb{M}_{z}^{\beta(1)} + \mathbb{M}_{z}^{\beta(2)} - \mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(3)} - 2\mathbb{T}_{xy}^{(5)}$$
$$T_{xxyx} = 2\mathbb{M}_{z}^{\beta(1)} - \mathbb{M}_{z}^{\beta(2)} - \mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(3)} + 2\mathbb{T}_{xy}^{(5)}$$
$$T_{zzxx} = 4\mathbb{M}_{xyz}^{(1)} + \mathbb{T}_{4v}^{(1)} + \mathbb{T}_{v}^{(1)} + \mathbb{T}_{v}^{(3)}$$
$$T_{zzxy} = -4\mathbb{M}_{z}^{\beta(1)} + 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + \mathbb{T}_{xy}^{(3)}$$
$$T_{yzyz} = \mathbb{M}_{xyz}^{(2)} - \mathbb{T}_{4v}^{(1)} - \mathbb{T}_{v}^{(2)} - \mathbb{T}_{v}^{(5)}$$
$$T_{yzzx} = -\mathbb{M}_{z}^{\beta(2)} + 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(2)} - \mathbb{T}_{xy}^{(5)}$$
$$T_{yzzy} = -\mathbb{M}_{xyz}^{(2)} - \mathbb{T}_{4v}^{(1)} - \mathbb{T}_{v}^{(2)} + \mathbb{T}_{v}^{(5)}$$
$$T_{yzxz} = \mathbb{M}_{z}^{\beta(2)} + 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(5)}$$
$$T_{xyxx} = -2\mathbb{M}_{z}^{\beta(1)} - \mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} - \mathbb{T}_{xy}^{(3)}$$
$$T_{xyzz} = 4\mathbb{M}_{z}^{\beta(1)} + 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} - \mathbb{T}_{xy}^{(3)}$$
$$T_{xyxy} = \mathbb{M}_{xyz}^{(2)} + 2\mathbb{T}_{v}^{(5)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & T_{yzyz} & T_{yzzx} & 0 & T_{yzzy} & T_{yzxz} & 0 \\ 0 & 0 & 0 & -T_{yzxz} & T_{yzzy} & 0 & -T_{yzzx} & T_{yzyz} & 0 \\ T_{xyxx} & -T_{xyxx} & 0 & 0 & 0 & T_{xyxy} & 0 & 0 & T_{xyxy} \end{bmatrix}$$

$$T_{yzyz} = \mathbb{M}_{xyz}^{(3)} + 2\mathbb{T}_{v}^{(4)} - \mathbb{T}_{v}^{(6)}$$
$$T_{yzzx} = \mathbb{M}_{z}^{\beta(3)} + 2\mathbb{T}_{xy}^{(4)} + \mathbb{T}_{xy}^{(6)}$$
$$T_{yzzy} = \mathbb{M}_{xyz}^{(3)} - 2\mathbb{T}_{v}^{(4)} - \mathbb{T}_{v}^{(6)}$$
$$T_{yzxz} = \mathbb{M}_{z}^{\beta(3)} - 2\mathbb{T}_{xy}^{(4)} + \mathbb{T}_{xy}^{(6)}$$
$$T_{xyxx} = \mathbb{M}_{z}^{\beta(3)} - 2\mathbb{T}_{xy}^{(6)}$$
$$T_{xyxy} = \mathbb{M}_{xyz}^{(3)} + 2\mathbb{T}_{v}^{(6)}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} T_{xxxx} & T_{xxxy} & 0 \\ T_{xxxy} & -T_{xxxx} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & T_{yyzz} \\ T_{zzxx} & T_{zzxy} & 0 \\ T_{xxyx} & -T_{yyzz} - T_{zzxx} & 0 \\ T_{yzzx} & T_{yzzy} & 0 \\ 0 & 0 & T_{zxxz} \\ -T_{yzzy} - T_{zxxz} & T_{xyyy} & 0 \\ 0 & 0 & -\frac{T_{xxxy}}{2} - \frac{T_{xxyx}}{2} + T_{xyyy} + T_{zzxy} \end{bmatrix}$$

$$T_{xxxx} = -\mathbb{T}_{4v}^{(1)} + 2\mathbb{T}_{v}^{(1)} + 4\mathbb{T}_{v}^{(2)}$$
$$T_{xxxy} = -\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(3)} - 2\mathbb{T}_{xy}^{(5)}$$
$$T_{yyzz} = -\mathbb{T}_{4v}^{(1)} - \mathbb{T}_{v}^{(1)} + \mathbb{T}_{v}^{(3)}$$
$$T_{zzxx} = \mathbb{T}_{4v}^{(1)} + \mathbb{T}_{v}^{(1)} + \mathbb{T}_{v}^{(3)}$$
$$T_{zzxy} = 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + \mathbb{T}_{xy}^{(3)}$$
$$T_{xxyx} = -\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(3)} + 2\mathbb{T}_{xy}^{(5)}$$
$$T_{yzzx} = 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(2)} - \mathbb{T}_{xy}^{(5)}$$
$$T_{yzzy} = -\mathbb{T}_{4v}^{(1)} - \mathbb{T}_{v}^{(2)} + \mathbb{T}_{v}^{(5)}$$
$$T_{zxxz} = \mathbb{T}_{4v}^{(1)} + \mathbb{T}_{v}^{(2)} + \mathbb{T}_{v}^{(5)}$$
$$T_{xyyy} = -\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} - \mathbb{T}_{xy}^{(3)}$$